FUNDAMENTALS OF AI AND ML

PROJECT

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TITLE: STUDENT PERFORMANCE PREDICTOR

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INTRODUCTION

In today’s academic environment, students often struggle to understand how their daily habits impact their performance. Factors such as study hours, attendance, and sleep patterns significantly influence academic outcomes, yet students rarely receive personalized insights based on these factors.

This project, Student Performance Predictor, is designed to bridge this gap by providing a simple and interactive platform where students can input their daily academic habits and receive predicted marks along with personalized improvement suggestions.

The project is built using Python and Streamlit, allowing real-time interaction and easy accessibility through a web interface.

PROBLEM STATEMENT

Key Problems:

- Lack of awareness about how study habits affect performance
- No real-time feedback system for students
- Poor balance between sleep, study, and attendance
- No personalized academic advice

This project aims to solve these issues by predicting performance and guiding students before exams.

OBJECTIVES & SCOPE

Objectives

The main objectives of this project are:

- To develop a system that predicts student performance based on key factors
- To provide a user-friendly web interface using Streamlit
- To ensure balanced evaluation (study + attendance + sleep)
- To generate personalized suggestions for improvement
- To help students make better academic decisions

Scope of the Project

This project is designed for:

- School and college students
- Self-assessment and improvement tracking
- Educational demonstrations

Future scope includes:

- Integration with real datasets
- Machine Learning-based predictions
- Performance tracking over time

METHODOLOGY

The project follows a simple rule-based prediction model instead of complex machine learning algorithms to maintain transparency and simplicity.

Steps followed:

1. User inputs data:
   - Name
   - Sleep hours
   - Study hours (per subject)
   - Attendance (per subject)
2. Data Processing:
   - Each subject is evaluated individually
   - A scoring formula is applied
3. Prediction Logic:
   - Study hours contribute significantly
   - Attendance contributes moderately
   - Sleep acts as a performance modifier
4. Output:
   - Predicted marks per subject
   - Overall average
   - Personalized advice

SYSTEM DESIGN

Input Parameters

- Sleep Hours (0–12)
- Study Hours (per subject)
- Attendance Percentage (per subject)

Processing Logic

- Base Score = (Study × weight) + (Attendance × weight)
- Sleep adjustment applied
- Penalty applied for imbalance

Output

- Subject-wise marks
- Overall performance
- Advice messages

ALGORITHM AND TECHONOLOGY USED

1. Take input from user
2. For each subject:
   - Calculate base score
   - Apply sleep modifier
   - Apply imbalance penalties
3. Limit marks to 100
4. Calculate average
5. Display results
6. Generate advice based on conditions

8. TECHNOLOGIES USED

- Python – Core programming language
- Streamlit – Web application framework
- VS Code – Development environment
- GitHub – Version control and project hosting

IMPLEMENTATION, RESULT & OUTPUT

IMPLEMENTATION

The project is implemented using Streamlit, which allows the creation of interactive web applications using Python.

Key Features:

- Multi-step user interface
- Interactive sliders for input
- Real-time prediction
- Conditional advice system
- Restart functionality

RESULT & OUTPUT

The system successfully:

- Predicts marks based on input data
- Maintains balance between study and attendance
- Prevents unrealistic predictions
- Provides actionable suggestions

Example Output:

- Subject-wise marks
- Average performance
- Advice like:
  - Increase study hours
  - Improve attendance
  - Maintain proper sleep

CHALLENGES & LEARNINGS

Challenges

- Setting up Streamlit environment
- Handling Python package issues (pip errors)
- Designing balanced prediction logic
- Avoiding unrealistic outputs
- Creating smooth user flow

Solutions

- Building real-world applications using Python
- Using Streamlit for web development
- Debugging environment and installation issues
- Designing logical algorithms
- Improving UI/UX for better interaction

CONCLUSION

Summary and Next Steps

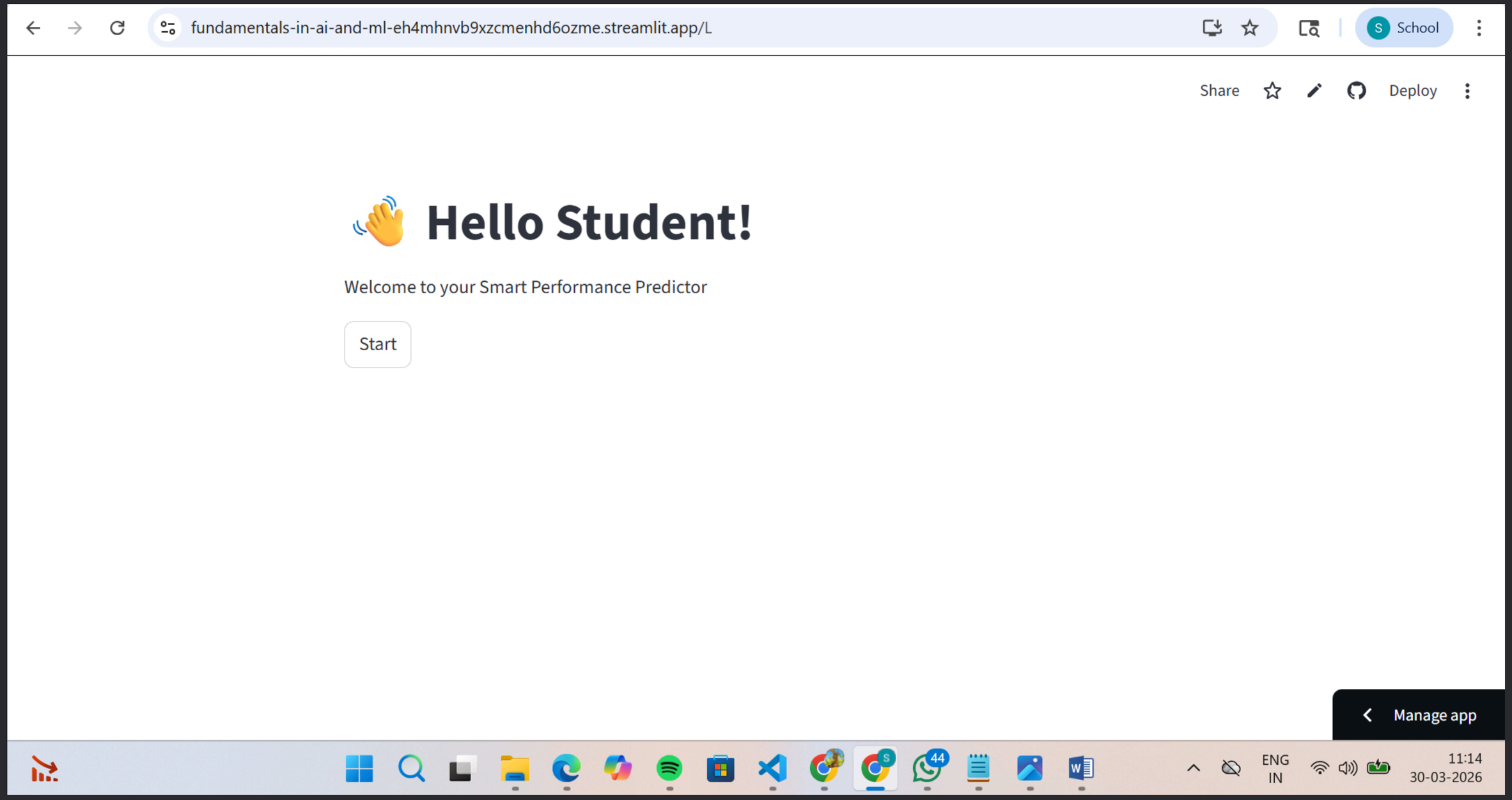
The Student Performance Predictor is a simple yet effective tool that helps students understand how their daily habits impact their academic performance. It promotes self-awareness and encourages better study practices.

The project demonstrates how basic programming concepts can be applied to solve real-world problems effectively.

Future Enhancements

- Machine Learning integration
- Graphical performance visualization
- Login system for students
- Data storage and history tracking
- Mobile-friendly interface

ABOUT THE APP



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School

Share Deploy

👩‍🎓 Enter Your Name

Your Name

SHRIYA

Next

Manage a

ENG IN

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Regarding your Hack
Vityarthi - Online Cou
Student Performance
Home - Canva
Student Performance
Ask Gemini
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📚 Hello SHRIYA, Enter Your Details

😴 Sleep Hours (per day)

6

Maths

Maths Study Hours

2

Maths Attendance (%)

75

Physics

Physics Study Hours

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Physics

Physics Study Hours

3

0 10

Physics Attendance (%)

75

Chemistry

Chemistry Study Hours

2

Chemistry Attendance (%)

75

Predict Performance

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📊 Your Performance Report

📈 Predicted Marks

Maths: 65

Physics: 65

Chemistry: 70

🎯 Overall Average: 66.67

💡 Advice

You're doing okay, but improvement is needed!

Restart

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